

Null Space of a Matrix

Definition

Null Space: The null space of a matrix $A \in \mathbb{R}^{m \times n}$ is defined to be

$$\text{Nul}(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}_m\}$$

Null Space Example

Let

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 6 \\ 3 & 1 & 2 \end{bmatrix}$$

Find $\text{Nul}(A)$. Solve $A\mathbf{x} = \mathbf{0}_3$.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 2 & 3 & 6 & 0 \\ 3 & 1 & 2 & 0 \end{array} \right] &\xrightarrow{R_2=R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 3 & 1 & 2 & 0 \end{array} \right] &\xrightarrow{R_3=R_3-3R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & -1 & -2 & 0 \end{array} \right] \\ &\xrightarrow{R_3=R_3+R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] &\xrightarrow{R_1=R_1-R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \mathbf{x} = \begin{bmatrix} 0 \\ -2t \\ t \end{bmatrix}, t \in \mathbb{R} \end{aligned}$$

Null Space is a Subspace.

Theorem

Let $A \in \mathbb{R}^{m \times n}$, then $\text{Nul}(A)$ is a subspace of $\mathbb{R}^n$.

Proof.

1. We know that $\mathbf{x} = \mathbf{0}$ is always a solution to $A\mathbf{x} = \mathbf{0}$
2. Let $\mathbf{u}, \mathbf{v} \in \text{Nul}(A)$, then see that

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

So, $\mathbf{u} + \mathbf{v} \in \text{Nul}(A)$ meaning it is closed under addition

3. Let $\mathbf{u} \in \text{Nul}(A)$, $c \in \mathbb{R}$. See that:

$$A(c\mathbf{u}) = cA(\mathbf{u}) = c\mathbf{0} = \mathbf{0}$$

So, $c\mathbf{u} \in \text{Nul}(A)$ meaning it is closed under multiplication.



Column Space of a Matrix

Definition

Column Space: The **column space** of a matrix $A = [\mathbf{a}_1 \ \dots \ \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ is denoted as $\text{Col}(A)$ and is the set of all linear combinations of columns of A .

$$\text{Col}(A) = \text{Span}(\mathbf{a}_1, \dots, \mathbf{a}_n) = \{\mathbf{b} \in \mathbb{R}^m \mid A\mathbf{x} = \mathbf{b} \text{ has a solution}\}$$

Finding a Basis of $\text{Col}(A)$ Example

Let

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 6 \\ 3 & 1 & 2 \end{bmatrix}$$

We want to know what an element of $\text{Col}(A)$ looks like, so we solve $A\mathbf{x} = \mathbf{b}$ and determine what $\mathbf{b}$ has to look like!

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 2 & 3 & 6 & b_2 \\ 3 & 1 & 2 & b_3 \end{array} \right] &\xrightarrow{R_2=R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 2 & b_2-2b_1 \\ 3 & 1 & 2 & b_3 \end{array} \right] &\xrightarrow{R_3=R_3-3R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 2 & b_2-2b_1 \\ 0 & -2 & -4 & b_3-3b_1 \end{array} \right] \\ &\xrightarrow{R_3=R_3+2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 2 & b_2-2b_1 \\ 0 & 0 & 0 & b_3+2b_2-7b_1 \end{array} \right] &\xrightarrow{R_1=R_1-R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3b_1-b_2 \\ 0 & 1 & 2 & b_2-2b_1 \\ 0 & 0 & 0 & b_3+2b_2-7b_1 \end{array} \right] \end{aligned}$$

Finding a Basis of $\text{Col}(A)$ Example Continued

So, all the systems that we can solve have the form of

$$\text{Col}(A) = \left\{ \mathbf{b} \in \mathbb{R}^3 \mid \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ 7b_1 - 2b_2 \end{bmatrix} \right\}$$

Which can be written as

$$b_1 \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix} + b_2 \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

Meaning

$$\begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

forms a basis of $\text{Col}(A)$

An Easier Way to Compute a Basis of $\text{Col}(A)$

1. Reduce to RREF
2. The columns of A corresponding to pivot columns form a basis of $\text{Col}(A)$.

Example

$$\begin{aligned} A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 6 \\ 3 & 1 & 2 \end{bmatrix} &\xrightarrow{R_2=R_2-2R_1} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 3 & 1 & 2 \end{bmatrix} \xrightarrow{R_3=R_3-3R_1} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{bmatrix} \xrightarrow{R_3=R_3+2R_2} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \\ &\xrightarrow{R_1=R_1-R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

So, the first two columns of A are a basis for $\text{Col}(A)$!

Showing These are Both Bases

We will show that

$$\text{Span} \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \right) = \text{Span} \left(\begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right)$$

$$\text{Span} \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \right) = \left\{ \mathbf{b} \in \mathbb{R}^3 \mid \mathbf{b} = \begin{bmatrix} c_1 + c_2 \\ 2c_1 + 3c_2 \\ 3c_1 + c_2 \end{bmatrix} \right\}$$

If we define $b_1 = c_1 + c_2$, $b_2 = 2c_1 + 3c_2$, and $b_3 = 3c_1 + c_2$, then

$$7b_1 - 2b_2 = 7(c_1 + c_2) - 2(2c_1 + 3c_2) = 7c_1 + 7c_2 - 4c_1 - 6c_2 = 3c_1 + c_2 = b_3$$

This is exactly what we said the systems we can solve look like!

The Column Space is a subspace of $\mathbb{R}^m$

We claim that for any $A \in \mathbb{R}^{m \times n}$ that $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$.

1. $A\mathbf{0}_n = \mathbf{0}_m$, so $\mathbf{0} \in \text{Col}(A)$.
2. Let $\mathbf{u}, \mathbf{v} \in \text{Col}(A)$. This means there are some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{u}$ and $A\mathbf{y} = \mathbf{v}$.
See that

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{u} + \mathbf{v}$$

So, $\mathbf{u} + \mathbf{v} \in \text{Col}(A)$.

3. Let $\mathbf{u} \in \text{Col}(A)$, $c \in \mathbb{R}$. Therefore, there is some $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{u}$. See that

$$A(c\mathbf{x}) = cA\mathbf{x} = c\mathbf{u}.$$

So, $c\mathbf{u} \in \text{Col}(A)$!

Col(A) and Nul(A) Practice

For the following matrix $A \in \mathbb{R}^{4 \times 3}$ find a basis for Col(A) and Nul(A).

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & 9 \\ 1 & 5 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & 9 \\ 1 & 5 & 7 \end{bmatrix} \xrightarrow[\substack{R_2=R_2-R_1 \\ R_3=R_3-2R_1}]{\substack{R_2=R_2-R_1 \\ R_3=R_3-2R_1}} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 3 & 3 \\ 1 & 5 & 7 \end{bmatrix} \xrightarrow{R_4=R_4-R_1} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 4 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Col(A) and Nul(A) Practice Continued

So, we have that

$$\text{Nul}(A) = \text{Span} \left(\begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} \right)$$

$$\text{Col}(A) = \text{Span} \left(\begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 5 \\ 5 \end{bmatrix} \right)$$

What do we notice about the dimension of these spaces?

Relating to Linear Transformations

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map, then we define:

Definition

Kernel: The **kernel** of T is the set of all $\mathbf{x} \in \mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{0}_m$.
This is just the null space of the matrix associated with T !

Definition

Image or **Range:** The **image** or **range** of T denoted

$$\text{Im}(T) = \text{Range}(T)$$

is the set of all $\mathbf{b} \in \mathbb{R}^m$ such that there is some $\mathbf{x} \in \mathbb{R}^n$ where

$$T(\mathbf{x}) = \mathbf{b}$$

This is just the column space of the matrix associated with T !

Row Space of a Matrix

Let $A \in \mathbb{R}^{m \times n}$, then we can write it as

$$A = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix}$$

Where each $r_k^\top \in \mathbb{R}^n$ is a **row** of A .

Note: We are transposing the rows to make them column vectors!

We define $\text{Row}(A)$ to be all linear combinations of the rows of A . Or:

$$\text{Row}(A) = \left\{ \mathbf{b} \in \mathbb{R}^n \mid \mathbf{b} = \sum_{k=1}^m c_k r_k^\top \right\} = \text{Col}(A^\top)$$

Rank of a Matrix

Definition

Rank: The **rank** of a matrix is the number of linearly independent rows and columns. This is also the number of pivots and the dimension of $\text{Col}(A)$!

Example

Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 6 \\ 3 & 1 & 2 \end{bmatrix}$, then we saw that we can row reduce to $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.

Which, has 2 pivots, so $\text{rank}(A) = 2$.

In addition, $\dim(\text{Col}(A)) = \dim(\text{Span}(\mathbf{a}_1, \mathbf{a}_2)) = 2$, so our definition is consistent!

Rank-Nullity Theorem

Theorem

Let $A \in \mathbb{R}^{m \times n}$, then we know that

$$\text{rank}(A) + \dim(\text{Nul}(A)) = n$$

$$\text{rank}(A) + \dim(\text{Nul}(A^\top)) = m$$